ASHESI UNIVERSITY

POST-HARVEST LOSS OF CEREAL PRODUCTS IN NKORANZA, BONO EAST REGION, GHANA

A Systemic and Theoretical Assessment of Post-Harvest Inefficiencies, Demographic Pressures, and Actionable Structural Interventions

Submitted in Partial Fulfillment of the Requirements for **SOAN 320: World Hunger, Population and Food Supplies** Department of Humanities and Social Sciences

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1. Introduction and Global Context for Post-Harvest Losses

Despite years of determined efforts to improve the production and quality standards of the world's food resources, food insecurity remains prevalent, particularly in nations of Africa and Southern Asia. By estimates on Sub-Saharan African Food Security, around one in three people faces severe malnourishment (Fraval et al., 2019). Considering that sub-Saharan Africa is a hotspot for food insecurity, it is evident that the issue remains unresolved, especially on a global scale.

Many studies have made their focus the issue of food insecurity. In the three most recent decades, in part due to the contributions of these studies and the support of world leaders, there have been general signs of turnaround. Most studies of late note the fact that global food production has far surpassed the threshold for population growth. (Foster & Leathers, 2017). These indicators, however, mask the severe disparities that exist in the sub-Saharan case of interest, for example.

According to the Food and Agriculture Organization (FAO), the International Food Policy Research Institute (IFPRI), and the World Food Program (WFP), food insecurity is not merely an issue of crop failure but also one of the poorly developed food systems and structures that make up the agricultural landscapes of food insecurity hotspots, similar to sub-Saharan Africa (Leathers & Leonard, 2023). Others may include increased dependence on food aid, limited success in adapting to climate change, inability to cope with high food prices, inefficient land use, and so on (Zorya et al., 2011).

In the sub-Saharan context, where, according to Abass et al. (2013), 70% of primary subsistence is determined by agriculture, food security is heavily constrained by low productivity and massive post-harvest losses (PHL) arising from structural inefficiencies in food systems and structures. At the same time, empirical research conclusions indicate that the reduction of PHL is far more cost-effective for sustained food security, despite traditional focus on farmland expansion and improving crop yields.

Nkoranza, Ghana, is a region that is highly representative of the conditions in which the aforementioned structural inefficiencies occur in a sub-Saharan African community. (Adjei & Boafo, 2019). It is dominated by smallholder farming families who mostly rely on rain-fed irrigation to support their economic activities. In addition, the dominant food products of this region are cereals such as maize, sorghum, and millet. (Adjei & Boafo, 2019; Nyebar et al., 2024). This report seeks to address the economic and nutritional challenges of PHL in Nkoranza as a model for food security improvement throughout sub-Saharan Africa.

2. Problem Definition, Iceberg Model, and Objectives

Smallholder farmers in Nkoranza struggle with post-harvest losses, which are driven by inadequate storage and limited access to preservation technology. This results in significant income loss, price volatility, and reduced food security.

We applied the Systems Iceberg model to study the underlying factors and beliefs that hold the visible problem.

At the **Events** layer, the immediate problem is grain spoilage after harvesting due to poor storage conditions, such as moisture and insect pests. MOFA identified inadequate storage

facilities and lack of ready markets as the two main causes of post-harvest losses (MOFA, 2011). As farmers lack storage, they may sell immediately after harvest, and this causes the price to fall. The Nkoranza North District Assembly also noted that buyers can take advantage of farmers by offering low prices because they lack storage facilities (Nkoranza North District Assembly, 2017). This contributes to decreasing household income and reduced food availability during the lean months.

The **Patterns** section shows how the problems persist over time. Grain prices usually drop around August and September, when farmers sell due to storage constraints, and rise around May when stocks are low. This recurring pattern demonstrates that post-harvest loss is not an isolated event but a persistent systemic problem.

Under **Structures**, the major issue is the limited availability of certified local warehouses and financing mechanisms. Without certified warehouses, farmers have limited access to warehouse receipt financing, while commercial banks generally view agricultural inventory as high-risk collateral (Nyebar et al., 2024). Although new warehouses were constructed in Nkoranza in 2021, the area had previously lacked adequate storage infrastructure (Ghana Districts, 2021). These interconnected barriers limit both farmers' ability to store produce and their access to capital.

For **Mental Models**, some deep beliefs held by stakeholders reinforce the system. Farmers may view informal storage and post-harvest losses as unavoidable, while financial institutions may perceive smallholder investment as too risky or unreliable to finance. Nkoranza may also be perceived as too remote to attract investment in storage infrastructure. These beliefs reinforce existing structural barriers and make the problem difficult to break.

Following the understanding of the problem’s root causes from the visible layer to the hidden structure assumptions that contribute to its complexity, our overall aim is to avoid post-harvest losses by ensuring appropriate harvest storage and providing financing for farmers’ income streams. We have identified two SMART objectives that serve as a guide for strategies in implementing this plan.

The first objective is to design and establish a short-term intervention plan to design and establish pre-harvest off-take contracts with GCX-approved buyers for at least three FBOs in Nkoranza before the next harvest season. The next major harvest in Nkoranza runs from March to June, giving ample time to plan and establish these contracts. In the first month, the team will conduct on-the-ground and online research to determine which FBOs are most suitable to work with, considering the number of farmers they serve and the feasibility of coordinating the contract. The FBOs with a low number of farmers are preferred to minimize bargaining costs. The next two months will be used to start the actual negotiation in the first month by identifying the needs of the buyers and the FBOs, and the second month to coordinate negotiation and finalize the contract draft. The contract will specify all agreements to be observed by both parties, including the amount of grain to be purchased and the price. In the remaining four months until harvest, the team will ensure that FBOs can provide the agreed-upon quantity of grain and advise on any challenges they face. The team will also plan the packaging and delivery logistics for buyers to ensure that all grain is delivered in good condition. Once the harvest is done, the team will ensure that buyers pay the FBOs and receive their share of the purchases.

With careful planning, this will be effective and ensure that all harvested grain is sold at a good market price and not lost.

The second objective is a long-term intervention plan to establish a Warehouse Receipt Financing (WRF) program in Nkoranza within about one year. For the program to be established, there should be certified warehouses that are accessible to farmers. Thus, we will identify a stakeholder who is willing to construct two warehouses, and that can be a Farmer-Based Organization (FBO), the government, or private companies, in 1-2 months. If we are unable to secure the construction of a warehouse, we will identify warehouses near Nkoranza and enquire about storage availability. If the two new warehouses, each capable of serving about 500 farmers, are constructed in 5-6 months, the next step is to contact the Ghana Commodity Exchange for warehouse certification within 1-2 months. Then this will be followed by establishing a formal agreement with at least one major agricultural financing institution, such as the Agricultural Development Bank (ADB), to provide short-term loans for farmers by accepting warehouse receipts as collateral. The short-term loan will provide farmers with an income stream during seasons when grain prices have fallen, helping them cover their expenses. Once that season ends, they can start selling their properly stored grains.

3. Stakeholder grid and group actors into clusters

Several actors are involved in the post-harvest loss of the cereal products in Nkoranza, and their actions and interests affect the effectiveness of the post-harvest operation in reducing cereal loss and food insecurity. The stakeholders can be categorized as follows: Farming Community, Market Actors, Government & NGOs, and Finance & Investment.

1. Farming community

Stakeholder	Role	Interest
Smallholder Farmers	Primary producers of Maize, cassava, millet, and sorghum are the major cereal crops.	Reduce post-harvest losses, increase the income, and improve food security
Farmer-based organizations or cooperatives (FBOs) (FAO, F. A. O. S. T. A. T. (2018)	Support collective action and coordination among Farmers.	Increase productivity, access to market, and profitability.
Farmers and supplies	Cultivate and supply cereal aggregators and markets.	Obtain fair prices and reduce grain losses to increase profitability.

2. Market Actors

Stakeholder	Role	Interest
Aggregators	Purchase and consolidate grain from farmers for the bulk sales.	Ensure a consistent supply of quality grain and reduce quality losses.
Market women	Transport and sell grain in Local and urban Markets	Access high-quality produce and minimize spoilage during transport and storage

Local community	Consumers and informal Market participants who depend on grain products.	Access affordable food and maintain stable food supplies.

3. Government and Non-government organizations

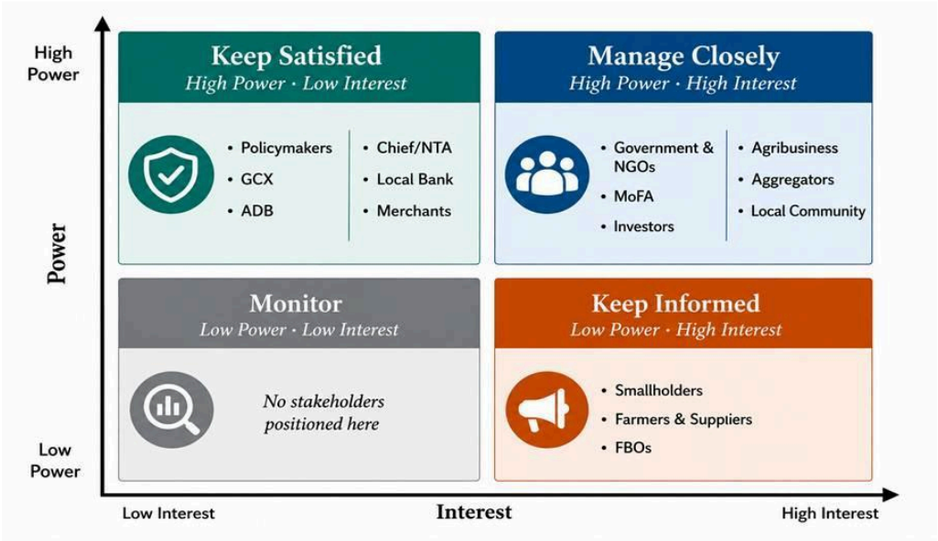
Stakeholder	Role	Interest
The Ministry of Food and Agriculture (MoFA).	Formulate policies in agriculture and offer extension services to the farmers.	Improve food security and reduce post-harvest losses
Nkoranza Tradition Authority (chief NTA)	Mobilize communities and support local development initiatives	Promote food security and reduce post-harvest losses
Council for Scientific and Industrial Research (CSIR)	Conduct research and develop post-harvest technologies and innovations.	Translate research findings into practical solutions for the farmers

4. Finance and Investment

Stakeholder	Role	Interest
Agriculture Development Bank (ADB)	Provide agriculture financing and specialized lending services	Support agricultural development while reducing financial risks.
Local Banks	Offer loan and credit facilities to farmers and traders	Improve farmers' profitability and ensure Loan repayment.
Ghana Commodity Exchange(GCX) (Adu-Sarkodee, G. (2020).	Provide structured trading systems and support warehouse receipt financing.	Improve market efficiency and reduce post-harvest losses.

The stakeholder map illustrates that a reduction of post-harvest cereal losses in Nkoranza depends on the collaboration of farmers, market actors, government institutions, and financial organizations. Grain losses directly impact farmers and FBOs, and a dependable supply of good-quality grain is critical to market actors. Government institutions facilitate policy support, research, and community mobilization, while financial institutions and market exchanges develop opportunities for storage, finance, and market structuring. These stakeholders are very instrumental in enhancing food security, farmers' livelihoods, and post-production losses in Nkoranza.

Stakeholder Matrix



Building on the stakeholder grid presented above, the power–interest matrix provides a more detailed basis for determining how each stakeholder can influence the implementation and success of the warehouse receipt pilot in Nkoranza. By classifying stakeholders by power and interests, the matrix enables the creation of engagement strategies tailored to the functions of each stakeholder, rather than a one-size-fits-all approach to stakeholder management. (Bernstein et al., 2020; Chiwala et al., 2025). This strategy is especially pertinent to the Nkoranza project, since the success of the warehouse receipt financing system hinges on cooperation along the entire cereal value chain. Farmers and farmer-based organizations are expected to supply and store cereals, while financial institutions provide access to finance, storage operators ensure appropriate grain handling, and buyers and other market actors support structured market access. The involvement and cooperation of these stakeholders can therefore significantly influence the implementation and sustainability of the pilot (Miranda et al., 2019; Safo, 2017).

Furthermore, the following table extends the Power versus Interest matrix, outlining tailored engagement strategies to manage institutional dynamics effectively.

Stakeholder Group	**Matrix Position**	**Engagement Strategy**
Local MFIs and the MoFA District Office	Manage Closely *(High Power - High Interest)*	Involved in planning, decision-making, financing arrangements, and regular project coordination.
District Assembly, Traditional Authority, and GCX	Keep Satisfied *(High Power - Low Interest)*	Maintain regular updates and consultation to secure continued institutional and market support
Smallholder farmers, farmers and suppliers, and FBOs	Keep Informed *(Low Power - High Interest)*	Provide regular information, training, and feedback on storage, financing, and market arrangements.

In Nkoranza's context, the proximate actors are smallholder farmers and FBOs, local aggregators and traders, ADB branches, and MFIs, as they directly engage with post-harvest losses, storage, and financing. Background actors include MoFA, District Assemblies, GCX, Traditional Authorities, among others, which influence the wider policy, regulatory, research, infrastructure, and institutional environment.

This classification aligns with Ghana's warehouse receipt literature, which highlights

farmers' limited access to credit and the importance of farmer organization and institutional support for effective warehouse receipt finance (Safo, 2017; Miranda et al., 2018; Miranda et al., 2019). For this reason, the financial institutions and district agriculture offices must be managed closely, while farmers should be kept informed through practical and accessible communication. Overall, the pilot depends on strong institutional collaboration and continuous farmer engagement to remain practical and trusted.

Theory, Literature, and Systems Analysis

This section provides a rigorous, academically grounded literature review and theoretical discussion of Nkoranza's post-harvest cereal loss, utilizing key economic, demographic, and agricultural concepts from the course materials (Foster & Leathers, 2017; Leathers & Leonard, 2023).

Critique of Malthusian Population Theory

The classical debate regarding global hunger is heavily rooted in the Malthusian population theory, which postulates that the human population grows exponentially (geometrically) while food supply can only grow linearly (arithmetically) (Foster & Leathers, 2017). Malthus argued that this divergence inevitably leads to a resource ceiling, triggering "positive checks" such as famine, disease, and war to reduce population back to the land's natural carrying capacity (Leathers & Leonard, 2023).

However, applying this classical framework to the Nkoranza North and South cereal basins reveals a major theoretical paradox (Adjei & Boafo, 2019; Kumar & Kalita, 2017). Nkoranza is highly productive and possesses significant biophysical potential to produce staple crops (Adjei & Boafo, 2019). The food deficit and seasonal hunger experienced by local households are not caused by an absolute biophysical limit on land production (Zorya et al., 2011). Instead, up to 60% of local cereal crops are lost after harvest due to technical inefficiencies, poor post-harvest handling, and lack of certified storage (Kumar & Kalita, 2017; Nyebar et al., 2024).

This empirical reality directly complicates Malthusian theory. If more than half of the physically produced food supply is lost to spoilage and pests, the food system is operating far below its biophysical limit (Kumar & Kalita, 2017). Localized hunger is not a natural result of land scarcity or overpopulation, but a man-made failure of technology and storage infrastructure. As Amartya Sen's entitlements approach highlights, food insecurity in Nkoranza is caused by a collapse of exchange entitlements, as farmers are forced to sell their crops under duress at extremely low prices, destroying their purchasing power (Foster & Leathers, 2017).

Demographic Transition Model and Cereal Demand

To understand the compounding demand for food in Ghana, we must analyze the national population structure through the Demographic Transition Model (DTM). The DTM outlines the transition of a country from high, fluctuating birth and death rates to low, stable rates as it develops economically (Foster & Leathers, 2017).

Ghana is currently in Stage 2/3 of the DTM, a phase of rapid population growth (Ghana Population Pyramid 2025). Death rates have fallen sharply due to improvements in healthcare, sanitation, and food supply, while birth rates remain elevated (Foster & Leathers, 2017).

Concurrently, Ghana's population structure is extremely youthful, with approximately 54.7% of the population under the age of 25 (Ghana Population Pyramid 2025). This youthful age profile creates powerful demographic momentum, ensuring that the national population will continue to expand rapidly for decades even as fertility rates decline (Foster & Leathers, 2017).

This rapid population expansion translates directly into a continuous rise in the aggregate demand for staple cereals, particularly maize, which provides more than 25% of the average Ghanaian household's caloric intake (Nyebar et al., 2024). In land-constrained environments, meeting this demand historically meant converting fragile ecosystems, forest reserves, or woodlands into cropland, leading to environmental degradation (Adjei & Boafo, 2019; Foster & Leathers, 2017).

In this context, reducing post-harvest losses behaves as a critical "land-saving technology" or "virtual land gain" (Zorya et al., 2011). By saving the grain that is already harvested, we can expand the net consumable food supply without bringing new land under cultivation (Zorya et al., 2011). Reducing Nkoranza's baseline post-harvest maize loss from 60% to a manageable 20% is equivalent to doubling the region's cultivated land area at zero marginal environmental and labor cost (Zorya et al., 2011).

Microeconomic Dynamics and Market Inelasticity

The financial vulnerability of smallholder farmers in Nkoranza is heavily driven by the microeconomic characteristics of staple crop demand (Nyebar et al., 2024). Food is a basic biological necessity, which means that the own-price elasticity of demand for cereals is highly inelastic, empirically estimated to range between -0.2 and -0.4 (Foster & Leathers, 2017). This own-price inelasticity indicates that consumers do not significantly alter their food consumption in response to changes in price (Foster & Leathers, 2017).

Because demand is highly inelastic, a sudden shift in the supply curve triggers massive price volatility (Foster & Leathers, 2017). During the major bumper harvest season in Nkoranza, the local supply of maize expands rapidly (Adjei & Boafo, 2019). Since smallholders lack storage facilities to hold this grain, they must sell it immediately at the farm-gate to secure cash (Zorya et al., 2011). This sudden supply shock shifts the short-run market supply curve to the right, causing farm-gate prices to crash dramatically (Adjei & Boafo, 2019; Nyebar et al., 2024).

Due to inelastic demand, the percentage drop in price is far larger than the percentage increase in quantity sold, meaning that the total revenues of the farming community actually decrease (Foster & Leathers, 2017). This economic reality is known as the "bumper harvest paradox" or King's Law (Foster & Leathers, 2017). Lacking access to certified storage or warehouse receipt financing, farmers are price-takers, completely vulnerable to predatory trading middlemen who exploit this structural vulnerability (Nyebar et al., 2024).

Technology, Purchased Inputs, and Environmental Enablers/Barriers

Agricultural intensification in Nkoranza is further analyzed through Ester Boserup's theory of agricultural growth, which challenges Malthusian pessimism by proposing that population growth is an independent variable that drives agricultural innovation (Boserup's Theory). As land becomes scarce, farmers are motivated to adopt purchased inputs, mechanical technologies, and intensive farming practices to maintain soil fertility and yields (Boserup's Theory; Foster &

Leathers, 2017). However, in Nkoranza's forest-savannah transition zone, smallholders face several critical enablers and barriers to technological intensification (Adjei & Boafo, 2019):

Subsidized Purchased Inputs: The Government of Ghana's agricultural policies, such as the "Planting for Food and Jobs" (PFJ) program, have acted as a yield enabler by providing subsidized certified seeds and nitrogen-based inorganic fertilizers (Nyebar et al., 2024). However, these fertilizer-intensive programs do not solve post-harvest bottlenecks and require continuous government financing to be sustainable.

Animal Traction and the Trypanosomiasis Barrier: While Boserup identifies animal-drawn plowing as a critical labor-saving technology (Boserup's Theory), this transition has faced a severe biological barrier in central Ghana (Animal traction). Nkoranza's humid transition zone is heavily infested with tsetse flies, the biological vector for animal African trypanosomiasis (AAT), or nagana (Animal traction; Alsan). Trypanosomiasis causes severe anemia, wasting, and high mortality in draft animals, historically excluding the use of oxen or horses in local farming (Animal traction; Alsan). This biological barrier has forced farmers to rely almost entirely on manual, high-effort human labor (representing 95% of agricultural draft power in Ghana), creating a major bottleneck for timely harvesting and land preparation (Animal traction).

Socio-Ecological Tensions: The push to increase cereal production has led to environmental degradation (Adjei & Boafo, 2019). The widespread use of manual slash-and-burn methods strips the soil, baring the land and exposing it to severe erosion and degradation (Adjei & Boafo, 2019). Furthermore, agricultural intensification contributes significantly to national greenhouse gas emissions (Ghana's Fifth National Greenhouse Gas Inventory). In Ghana, agriculture and land-use change are major sources of emissions, driven by carbon dioxide (CO_2) released during deforestation, methane (CH_4) emitted from flooded rice paddies and enteric fermentation in livestock, and nitrous oxide (N_2O) from synthetic nitrogen fertilizers (Ghana's Fifth National Greenhouse Gas Inventory). This creates a direct tension between agricultural output expansion and climate mitigation goals (Foster & Leathers, 2017; Ghana's Fifth National Greenhouse Gas Inventory).

Actionable Priorities and Sustainability Matrix

To help stop crop losses after harvest in Nkoranza, we have created three main goals. Since our team consists of busy university students, these actions are designed to be quick, completely online, and require zero physical travel.

Priority 1: Secondary-Data-Driven Research and Assessment

This priority focuses on establishing a robust empirical baseline to guide local agricultural policy. Instead of conducting resource-intensive and physically demanding field surveys, the research team will design a desk-study methodology that leverages secondary data from the Ministry of Food and Agriculture regional crop yield databases (Rethinking Agricultural Development in Sub-Saharan Africa). To capture critical qualitative insights from Nkoranza's agricultural community, group members will execute a targeted, phone-based SMS and WhatsApp survey of registered Farmer-Based Organization leaders. This digital survey will evaluate local storage awareness, trace the prevalence of grain weevils and Fall Armyworm

damage, and assess mycotoxin and aflatoxin risk perceptions among FBO members. This secondary and digital methodology eliminates travel requirements while gathering precise, actionable data that can be directly integrated into university capstone projects and District Assembly policy frameworks.

Priority 2: Zero-Inventory Digital Brokerage Agency

This priority designs a digital entrepreneurship initiative to resolve market access and pricing challenges. Group members will establish a zero-inventory, commission-based digital agency using a standard WhatsApp Business directory and a centralized SMS database (Rethinking Agricultural Development in Sub-Saharan Africa). This digital brokerage service will connect local cereal FBOs in Nkoranza directly with already-existing, GCX-certified aggregators, processors, and secure storage warehouses operating in the region. By acting as a digital intermediary, the brokerage will charge a minor commission on bulk-sale transactions and storage space bookings, while providing smallholders with real-time price updates and demand forecasts. This business model requires zero physical inventory, no warehouse construction capital, and no travel, leveraging existing digital infrastructure to eliminate market information asymmetries and protect smallholders from predatory middlemen.

Priority 3: Digital Policy Advocacy Campaign

This priority outlines a targeted policy intervention to mobilize public resources and municipal support for post-harvest infrastructure. Rather than engaging in slow, physical administrative lobbying, group members will draft a standardized, professional Policy Brief template that can be emailed directly to the Nkoranza North and South District Assemblies, MoFA directorates, and local agricultural officers (Rethinking Agricultural Development in Sub-Saharan Africa). The Policy Brief will advocate for municipal co-financing of community-level storage silos, and will be accompanied by a pre-designed social media advocacy toolkit. This digital toolkit, featuring pre-written graphics and hashtags, will be shared across student networks and youth-led civic groups to generate local public interest and mobilize civic pressure. This zero-travel advocacy model utilizes targeted, professional communication to influence local policy and push the public sector to support decentralized grain storage infrastructure.

Sustainability Matrix

To evaluate the viability of these three low-effort actions, they are passed through a sustainability matrix, evaluating them across four distinct lenses in the Table

Table 2: Sustainability Matrix of Student-Led Actionable Priorities

Priority Action	**Economic Lens**	**Social Lens**	**Environmental Lens**	**Institutional Lens**
Research Priority: Secondary Yield & FBO Digital Survey	High Sustainability: Requires near-zero financial resources, utilizing academic internet access.	High Sustainability: Builds local data capacity and respects FBO time via phone surveys.	High Sustainability: Zero carbon footprint from zero physical travel.	High Sustainability: Easily integrated into Ashesi's existing research frameworks.
Entrepreneurship: WhatsApp Brokerage Directory	High Sustainability: Low overhead; generates commission	High Sustainability: Empowers smallholders by eliminating trader	High Sustainability: Low footprint; optimizes truck	Medium Sustainability: Depends on the performance and

	revenues from transaction fees.	information asymmetries.	routing to reduce transport emissions.	capacity of external GCX systems.
Policy Priority: Digital Policy Brief & Advocacy	High Sustainability: Extremely cost-effective, utilizing email and social networks.	High Sustainability: Engages youth in civic advocacy for community development.	High Sustainability: Neutral; directly advocates for climate-smart storage solutions.	Medium Sustainability: Relies on the responsiveness of municipal bureaucratic structures.

The reasoning for these scores is detailed below in full prose:

Economic Sustainability (Money):
Excellent. Because everything is done online, we don't need a lot of money to start, which removes financial risk. The digital brokerage will even make a small profit to keep itself running.

Social Sustainability(People):
Excellent. Our plan empowers farmers by giving them direct market information, cutting out unfair middlemen. The online campaign also gets young people actively involved in improving their community.

Environmental Sustainability(Nature):
Excellent. Since we aren't traveling, building new structures, or clearing land, our carbon footprint is zero. By making better use of existing storage, we also help protect land and prevent deforestation.

Institutional Sustainability(Partnership):
Good. The research fits perfectly with our university work. The other projects rely on existing government and market networks, though their ultimate success will partly depend on how quickly local government offices reply to our emails.

Community Job Creation Impact

Collectively, these three digital priorities will stimulate new forms of employment and income generation within Nkoranza's agricultural ecosystem. The **Secondary-Data Research** initiative will create short-term research assistant roles for university students and local data collectors supporting phone-based surveys. The **Digital Brokerage Agency** will generate ongoing opportunities for youth entrepreneurs to manage WhatsApp directories, coordinate transactions, and provide digital marketing services to FBOs and warehouse operators. Meanwhile, the **Policy Advocacy Campaign** will engage communication specialists, graphic designers, and social media coordinators in producing advocacy materials and managing online outreach. Together, these interventions promote inclusive, skill-based employment that leverages digital literacy rather than physical labor, empowering young people to participate in Ghana's evolving agri-tech economy.

Conclusion

Finally, the food insecurity in Nkoranza is not only due to inadequate food production but also to cereal losses post-harvest. Despite the significance of Nkoranza as a cereal-producing area, smallholder farmers are still losing out and experiencing food insecurity during the seasons due to a lack of storage facilities, access to credit facilities, volatility of prices, and poor post-production systems. The issue is actually about structural and institutional shortcomings of

the agricultural value chain, not about the lack of productive capacity.

The theoretical analysis also indicates that the case of Nkoranza calls into question the Malthusian scenario of hunger. While there are significant pressures on population growth and growing cereal demand, there is evidence that enhancements in the availability of existing food supplies, through storage, financing, and distribution systems, can lead to more food available without the need to simply expand agricultural land. The analysis of the DT Model, microeconomic price inelasticity, and Boserup's theory also underscore the role of population, market conditions, technology, and innovation in agriculture in determining food security.

To overcome these challenges, the report recommends two interventions, namely, pre-harvest off-take contracts with the GCX-approved buyers and the Warehouse Receipt Financing system. These interventions can collectively help to lower farmers' need for an on-farm cash sale, better access to credit facilities, lower price volatility, and safer grain storage. However, their success relies on the successful cooperation between the farmers and the FBOs, market actors, government institutions, GCX, and financial institutions.

Secondary data research, the zero-inventory digital brokerage agency, and digital policy advocacy are low-cost and practical approaches to supporting the broader intervention. They are economically, socially, and environmentally sustainable, which shows that it is possible to contribute to food security without investing large amounts of money or depending on large physical investments. These activities can also open the way for youth employment and digital entrepreneurship in the agricultural system of Nkoranza.

In the end, minimizing post-harvest losses involves making more effective use of food that is already being produced. Nkoranza has the potential to enhance farmer livelihoods, food availability, and environmental stress, through strengthening storage, financing, market access, and stakeholder partnerships, as well as digital innovation. Therefore, the remedy of the post-harvest losses should be considered as a food security problem that needs a concerted and sustainable effort and a systemic approach to the problem.

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